

# WestQuant Representation Scheduler (Project A)

## Slide 1 — Project Overview

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**Problem:** A fixed logical Hamiltonian does not determine a unique useful quantum representation. The physical embedding/layout matters.

**Approach:** Automated search over QoolQit embeddings (InteractionEmbedder, SpringLayoutEmbedder, Blade) using successive halving across

**Results:** Flagship experiment (6 problems, 9 embeddings, 3 replicates): embedding main effect  $\eta^2(R) = 19.0\%$  (median). For 5/6 problems,

**Importance:** Treating the embedding as an optimization variable (not a fixed choice) materially improves quantum solution quality. Open-so

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## Slide 2 — QoolQit Experience

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**Positive aspects:** QoolQit's Register, Drive, and QuantumProgram abstractions are clean and composable. The embedding API (Interaction

**Challenges:** DMM drives require a DMM-capable device (AnalogDeviceWithDMM, not AnalogDevice). The max\_energy compilation profile is

**Workflow:** The successive-halving approach (cheap metrics first, expensive emulation last) is essential for scaling the search. QoolQit's inte